Crack Width Verification - RC Slab (EN 1992-1-1 7.3.4)

Characteristic crack width for a reinforced concrete slab under quasi-permanent loading.

1. Input Parameters

Symbol	Value	Unit	Description
b	1000	mm	Section width (per metre strip)
h	300	mm	Section depth
d	260	mm	Effective depth
c_{nom}	30	mm	Nominal concrete cover
ϕ	16	mm	Bar diameter
A_s	1005	mm^2	Tension reinforcement area (5 H16)
M_{qp}	45	kNm	Quasi-permanent bending moment
E_s	200000	MPa	Steel modulus of elasticity
E_{cm}	33000	MPa	Concrete secant modulus
$f_{ct,eff}$	2.9	MPa	Effective concrete tensile strength
k_t	0.4	-	Load duration factor (long term)
k_1	0.8	-	Bond factor (high-bond bars)
k_2	0.5	-	Strain distribution factor (bending)
k_3	3.4	-	Code coefficient
k_4	0.425	-	Code coefficient
w_{max}	0.3	mm	Allowable crack width (XC exposure)

2. Calculation

Modular ratio E_s/E_cm

$$\alpha_e = \frac{E_s}{E_{cm}}$$
$$= \frac{200000}{33000}$$
$$= \mathbf{6.061}$$

Neutral axis depth (cracked section)

$$x = \frac{A_s \cdot \alpha_e \cdot \left(\sqrt{\frac{2 \cdot b \cdot d}{A_s \cdot \alpha_e} + 1} - 1\right)}{b}$$
$$= \frac{1005 \cdot 6.061 \cdot \left(\sqrt{\frac{2 \cdot 1000 \cdot 260}{1005 \cdot 6.061} + 1} - 1\right)}{1000}$$
$$= \mathbf{50.52}\ \text{mm}$$

Internal lever arm

$$z = d - \frac{x}{3}$$
$$= 260 - \frac{50.52}{3}$$
$$= \mathbf{243.2}\ \text{mm}$$

Steel stress under quasi-permanent loads

$$\sigma_s = M_{qp} \cdot 1000000 \cdot \frac{1}{A_s \cdot z}$$
$$= 45 \cdot 1000000 \cdot \frac{1}{1005 \cdot 243.2}$$
$$= \mathbf{184.1} \text{ MPa}$$

Effective tension zone height

$$h_{c,eff} = \min\left(\frac{h}{2}, \frac{h-x}{3}, 2.5 \cdot (-d+h)\right)$$
$$= \min\left(\frac{300}{2}, \frac{300-50.52}{3}, 2.5 \cdot (-260+300)\right)$$
$$= \mathbf{83.16} \text{ mm}$$

Effective reinforcement ratio

$$\rho_{p,eff} = \frac{A_s}{b \cdot h_{c,eff}}$$
$$= \frac{1005}{1000 \cdot 83.16}$$
$$= \mathbf{0.01208}$$

Maximum crack spacing

$$s_{r,max} = c_{nom} \cdot k_3 + \frac{k_1 \cdot k_2 \cdot k_4 \cdot \phi}{\rho_{p,eff}}$$
$$= 3.4 \cdot 30 + \frac{0.425 \cdot 0.5 \cdot 0.8 \cdot 16}{0.01208}$$
$$= \mathbf{327.1} \text{ mm}$$

Mean strain difference (eps_sm - eps_cm)

$$\varepsilon_{diff} = \max\left(\frac{0.6 \cdot \sigma_s}{E_s}, \frac{-\frac{f_{ct,eff} \cdot k_t \cdot (\alpha_e \cdot \rho_{p,eff}+1)}{\rho_{p,eff}} + \sigma_s}{E_s}\right)$$
$$= \max\left(\frac{0.6 \cdot 184.1}{200000}, \frac{184.1 - \frac{0.4 \cdot 2.9 \cdot (0.01208 \cdot 6.061 + 1)}{0.01208}}{200000}\right)$$
$$= \mathbf{0.0005524}$$

Characteristic crack width

$$w_k = \varepsilon_{diff} \cdot s_{r,max}$$
$$= 0.0005524 \cdot 327.1$$
$$= \mathbf{0.1807} \text{ mm}$$

3. Verification

OK $w_k = 0.1807 \text{ mm} \le w_{max} = 0.3 \text{ mm}$